SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717- SOFTWARE TESTING

EPX:1

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Identification and Documentation of Software Defects in an ATM (Automated Teller Machine) System

Aim

To study the specifications of an ATM system, test it with valid and invalid inputs, and identify, classify and document the software defects found, using a standard defect report format.

Problem Statement

An ATM system lets bank customers insert a card, verify identity with a PIN, check balances, withdraw cash, change their PIN and receive a receipt. The system handles money and confidential data, so even a small defect can cause financial loss, security breaches or loss of customer trust.

Verification is needed to check that the ATM:

- Behaves correctly for valid inputs.
- Rejects invalid inputs safely.
- Meets every stated requirement.

Any deviation from the specification must be recorded as a defect, with enough detail for developers to reproduce and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	The system must accept only a valid, readable bank card.
FR2	The user must enter a 4-digit PIN. The PIN must be masked on screen.
FR3	The card must be blocked after 3 consecutive incorrect PIN attempts.
FR4	The user can check their account balance.
FR5	Withdrawals must be in multiples of ₹100, with a minimum of ₹100 and a maximum of ₹20,000 per transaction.
FR6	The daily withdrawal limit is ₹40,000 per card.
FR7	A withdrawal must not exceed the account balance or the cash available in the machine.
FR8	The account must be debited only if cash is dispensed successfully.
FR9	The session must end automatically after 30 seconds of inactivity, and the card must be returned.

ID	Requirement
FR10	The receipt must show the correct transaction amount and available balance.
FR11	The system must show a clear, appropriate error message for every invalid action.
FR12	The user can change their PIN. The new PIN must differ from the old PIN.

2 Software and environment requirements

Item	Details
ATM application	Java/C# based ATM simulator (or a bank ATM test build)
Database	MySQL / Oracle (customer accounts and transaction logs)
Operating system	Windows 10/11 or Linux
Testing tools	Test case document (Excel), bug tracker (Jira / Bugzilla)
Test data	Valid and invalid cards, correct and incorrect PINs, various balances and amounts

Defect Report

1 Defect summary

Defect ID	Module	Defect Title	Violated Req.	Severity	Priority	Status
ATM-D01	Authentication	Card not blocked after 3 wrong PIN attempts	FR3	Critical	High	Open
ATM-D02	Withdrawal	Amount not a multiple of ₹100 is accepted	FR5	High	High	Open
ATM-D03	Withdrawal	Account debited although cash was not dispensed	FR8	Critical	High	Open
ATM-D04	Withdrawal	Daily limit not enforced across multiple transactions	FR6	High	High	Open
ATM-D05	Session	No automatic timeout; card not returned	FR9	Medium	Medium	Open
ATM-	PIN	PIN shown in plain text	FR2	High	High	Open

Defect ID	Module	Defect Title	Violated Req.	Severity	Priority	Status
D06	Change	screen				
ATM-D07	PIN Change	New PIN accepted even if same as old PIN	FR12	Medium	Medium	Open
ATM-D08	Receipt	Receipt shows balance before deduction	FR10	Medium	Medium	Open
ATM-D09	Error handling	Wrong message shown for insufficient balance	FR11	Low	Low	Open

2 Detailed defect reports

Defect ID: ATM-D01

- **Title:** Card is not blocked after three consecutive incorrect PIN entries
- **Module:** Authentication
- **Severity / Priority:** Critical / High
- **Precondition:** A valid card is inserted.
- **Steps to reproduce:**
    a. Insert a valid card.
    b. Enter an incorrect PIN (e.g., 1111).
    c. Repeat with incorrect PINs three more times.
- **Expected result:** The card is blocked after the 3rd wrong attempt, with the message "Card blocked. Contact your bank."
- **Actual result:** The system keeps allowing PIN attempts. A 4th, 5th and later attempts are accepted.
- **Impact:** An unauthorized person can guess the PIN by brute force, which is a serious security risk.
- **Suggested fix:** Keep a failed-attempt counter per card in the database, and block the card when it reaches 3.
- **Status:** Open

Defect ID: ATM-D02

- **Title:** Withdrawal amount that is not a multiple of ₹100 is accepted
- **Module:** Cash Withdrawal
- **Severity / Priority:** High / High
- **Steps to reproduce:**
    a. Log in with a valid card and PIN.
    b. Select Withdraw.
    c. Enter ₹750 and confirm.
- **Expected result:** The request is rejected with "Enter an amount in multiples of ₹100."
- **Actual result:** The transaction proceeds, and the machine tries to dispense ₹750.

- **Impact:** The machine cannot physically dispense this amount with ₹100 notes. This can cause a dispenser error or an incorrect debit.
- **Suggested fix:** Add the validation `amount % 100 == 0` before processing.
- **Status:** Open

Defect ID: ATM-D03

- **Title:** Account is debited even though cash was not dispensed
- **Module:** Cash Withdrawal / Hardware interface
- **Severity / Priority:** Critical / High
- **Steps to reproduce:**
    a. Log in and request a withdrawal of ₹2,000.
    b. Simulate a cash dispenser failure (e.g., note jam).
    c. Check the account balance.
- **Expected result:** The transaction is cancelled and the account is not debited.
- **Actual result:** ₹2,000 is deducted, and the screen shows "Technical error".
- **Impact:** The customer loses money, and the bank gets a dispute and compliance issue.
- **Suggested fix:** Use transaction handling. Commit the debit only after the dispenser confirms success, and roll back otherwise.
- **Status:** Open

Defect ID: ATM-D04

- **Title:** Daily withdrawal limit is not enforced
- **Module:** Cash Withdrawal
- **Severity / Priority:** High / High
- **Steps to reproduce:**
    a. Withdraw ₹20,000 twice (total ₹40,000).
    b. Start a new session and request ₹10,000.
- **Expected result:** Rejected with "Daily limit exceeded."
- **Actual result:** The ₹10,000 withdrawal is allowed. The system checks only the per-transaction limit.
- **Impact:** The limit meant to reduce fraud loss on lost or stolen cards does not work.
- **Suggested fix:** Sum the day's withdrawals for the card and check this total before approving.
- **Status:** Open

Defect ID: ATM-D06

- **Title:** PIN is displayed in plain text on the PIN change screen
- **Module:** PIN Change
- **Severity / Priority:** High / High
- **Steps to reproduce:**
    a. Log in and select Change PIN.
    b. Type the new PIN.
- **Expected result:** Digits are masked (****).

- **Actual result:** The digits appear in plain text.
- **Impact:** Shoulder surfing can expose the PIN, which is a confidentiality breach.
- **Suggested fix:** Use a password-type input field to mask all PIN entries.
- **Status:** Open

The remaining defects (D05, D07, D08, D09) can be written in the same format.

3 Specification-level defects:

Reviewing the specification itself also shows gaps that could cause defects later:

- It does not say what happens if power or network fails mid-transaction.
- It does not say when the "daily limit" resets, whether at midnight or after 24 hours.
- The minimum and maximum PIN-change attempts are not defined.
- It does not specify the behaviour for an expired or damaged card.

4 Defect life cycle:

New → Assigned → In Progress → Fixed → Retesting → Closed. If the defect persists after retesting, it is reopened and sent back to the developer.

Result / Conclusion:

Result: Testing the ATM system against its specification found 9 defects:

Severity	Count
Critical	2
High	4
Medium	2
Low	1

Most defects were in the **Authentication** and **Cash Withdrawal** modules, which are the most security- and money-critical parts. Four defects (D01, D03, D04, D06) can cause financial loss or security risks.

Conclusion:

The ATM system does not yet meet all of its stated requirements and should not be released until the Critical and High defects are fixed and retested. The case study shows that comparing the system against its specification, testing with both valid and invalid inputs, and using a structured defect report helps find defects early and gives developers what they need to fix them. Thorough testing is essential for safety-critical financial software like an ATM.